Too Long, Didn't Model: Decomposing LLM Long-Context Understanding With Novels

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Abstract

Although the context length of large language models (LLMs) has increased to millions of tokens, evaluating their effectiveness beyond needle-in-a-haystack approaches has proven difficult. We argue that novels provide a case study of subtle, complicated structure and long-range semantic dependencies often over 128k tokens in length. Inspired by work on computational novel analysis, we release the Too Long, Didn't Model (TLDM) benchmark, which tests a model's ability to report plot summary, storyworld configuration, and elapsed narrative time. We find that none of seven tested frontier LLMs retain stable understanding beyond 64k tokens. Our results suggest language model developers must look beyond "lost in the middle" benchmarks when evaluating model performance in complex long-context scenarios. To aid in further development we release the TLDM benchmark together with reference code and data.

1 Introduction

Large language model (LLM) context lengths have expanded to millions of tokens, theoretically enabling the analysis of long and complicated documents. However, recent research suggests LLMs do not properly integrate information across long contexts (Hsieh et al., 2024; Karpinska et al., 2024) and have difficulty keeping track of order within contexts (Merrill et al., 2024; Li et al., 2025). This failure mode has been hard to evaluate.

Current long-context benchmarks like Needle In a Haystack (Kamradt, 2023) and Passkey Retrieval (Mohtashami and Jaggi, 2023) evaluate a model's minimal ability to access "lost in the middle" data, but fail to test long-context *understanding*. Retrieving one relevant document from a sea of irrelevant documents does not replicate how we expect most users use long contexts: to integrate multiple documents to arrive at complex conclusions.

We therefore present the Too Long, Didn't Model (TLDM) benchmark: a pipeline for testing long-context LLMs on a set of forty English-language novels ranging in length from <32k to >128k tokens using a suite of narrative understanding tasks including summarization, storyworld description, and narrative time estimation. Our benchmark circumvents limitations imposed by the need for human annotations by selecting narrative tasks suitable for associative ground construction. For example, a novel summary can be decomposed to a series of concatenated chapter summaries. This property allows us to assess model stability in long-context regimes by measuring the difference between novel-level and chapter-level predictions.[1]

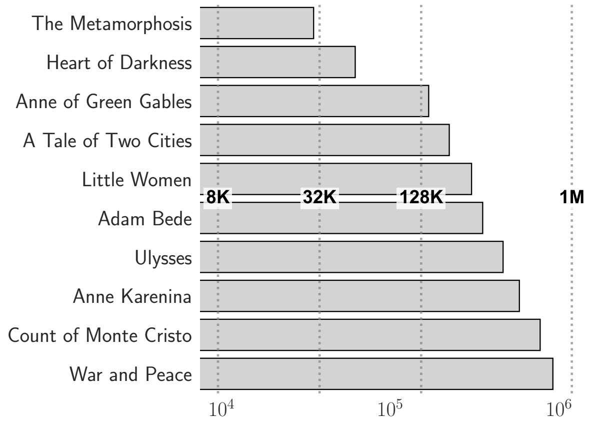

Figure 1: Token lengths of ten popular novels as tokenized by Gemma 3 contrasted with common maximum LLM context lengths (log) as indicated by dotted lines.

2 Related Work

State sequencing. It is unknown whether self-supervised machine learning models are able to track state over long inputs. While theory suggests they can (Liu et al., 2023; Merrill and Sabharwal, 2025), certain models fail to do so in practice. Merrill et al. (2024) show state-space models (e.g. S4 and Mamba) struggle with state-tracking tasks. Transformers fare better: researchers have found

[1]We make our data and code available here.

empirical evidence for state tracking in tasks like entity tracking (Li et al., 2021; Kim and Schuster, 2023), permutation composition (Li et al., 2025), Othello (Li et al., 2023), and chess (Karvonen, 2024). These results indicate transformers can theoretically process long-context input.

Long-context benchmarking. It is popular to test frontier transformers on long-context inputs by subjecting them to benchmarks such as Needle in a Haystack (Kamradt, 2023) and passkey retrieval (Mohtashami and Jaggi, 2023). These benchmarks test long-context processing by having the model retrieve relevant documents randomly shuffled into sets of irrelevant documents. Early long-context models (e.g. GPT-4) performed poorly on these tasks, which encouraged language model developers to forefront long-context testing (Liu et al., 2024; Li et al., 2024; Zhang et al., 2024).

Literature as benchmark. Competent "lost in the middle" performance does not mean models can integrate information over long contexts. Researchers have therefore proposed benchmarks for assessing models on more complicated tasks such as question answering (Wang et al., 2024; Yuan et al., 2024), "multi-hop reasoning" (Roberts et al., 2024), instruction following (Bai et al., 2024), or all of the above (Chen et al., 2025; Hsieh et al., 2024). Benchmarks have likewise begun to turn to literature as a potential source of natural long-context data (Sun et al., 2021; Kim et al., 2024; Ahuja et al., 2025). One such benchmark, NoCha (Karpinska et al., 2024) finds open-weight models achieve only near-random accuracy when querying texts averaging 127k tokens in length — but NoCha (and other literary benchmarks) fail to provide insight into *when* in the context window models begin to fail to process information. Our benchmark proposes a unique set of state-oriented narrative understanding tasks to test how models represent state *up to a given point in a novel*.

3 Methods

Data. We align Project Gutenberg's catalog (Hart, 1971) with the MultiHATHI corpus (Hamilton and Piper, 2023) to identify 6,219 English-only public domain works of fiction averaging 241 pages in length. We then randomly sample books containing clearly delineated chapters that appear to follow a single narrative. We keep the first ten suitable texts that fall into each of four length bins: <32k, 32k–64k, 64k–128k, and >128k tokens.[2,3] Our final dataset contains forty novels containing an average of 27 chapters each.

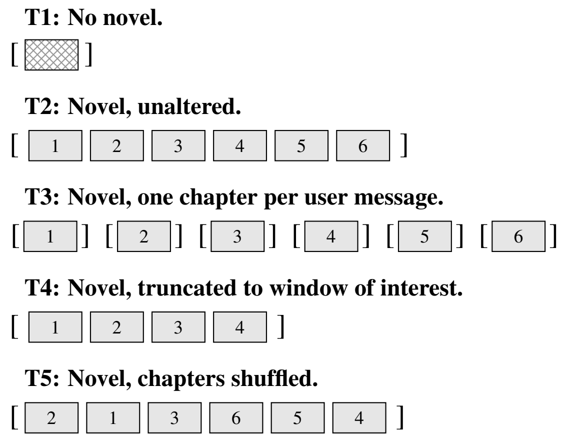

Figure 2: The five treatments considered in our study as applied to a six-chapter novel. Brackets indicate user message beginning and end while rectangles indicate chapters. Crosshatches indicate absence of input.

Tasks. We deploy three narrative understanding tasks that require processing large amounts of text:

1. **Summarization:** Summarize the narrative with one sentence per chapter.

2. **Storyworld description:** Return the last known physical location of every character in the narrative.

3. **Narrative time:** Estimate the narrative time passed in hours, days, months, or years.

Each task requires models to extract and report different types of narrative information; the first requires identifying salient plot points, the second requires entity tracking, and the third requires a model of story time independent of narration.[4]

Windows of interest. Each task-specific prompt instructs the model to perform the task for the first 25%, 50%, 75%, or 100% of chapters. This allows us to evaluate whether models can limit results to a subsection of text. Subsequent results will refer to these subspans as *windows of interest.*

Text treatments. We probe for the circumstances in which LLMs fail to process long contexts by permuting all texts with five treatments, presented in

[2]The length of each text in number of tokens is calculated by applying the Gemma 2 tokenizer to the text file versions of each text with paratextual information removed.

[3]We list all sampled novels in Appendix A.

[4]Task-specific prompts are made available in the appendix.

Figure 2.[5] Our first treatment (T1) forgoes the actual text for the novel title and author.[6] Our second treatment (T2) passes in the input text unaltered. The third treatment (T3) wraps each chapter in a unique user message, with the intuition being that explicitly delineating chapters could aid models in parsing long inputs. Our fourth treatment (T4) truncates the input text to the window of interest. Our final treatment (T5) randomly shuffles the chapters to test whether models are able to reconstruct narratives from anachronous input text. We further consider all possible combinations of T3-5 for a total of nine treatments over each input text.

Evaluation. TLDM assesses the stability of model predictions made over long contexts relative to short-context responses. We do not have human-labeled ground truth, but instead compare individual model performance on short contexts to performance on long contexts. Contemporary LLMs are often pretrained with a context length of 4096 tokens before being generalized to longer contexts in post-training (Abdin et al., 2024; Yang et al., 2025; Su et al., 2023). This token range is approximately the length of the average English novel chapter.[7] We therefore generate chapter-level outputs independently and concatenate them to create model-specific novel-level predictions.[8] We then prompt the model with each text treated as described in section 3. We finally compute the difference between these full-text predictions and the concatenated short-context output using a similarity heuristic normalized to the range $[0, 1]$.[9]

4 Results

We test seven recent frontier models (Table 1) on the TLDM benchmark to evaluate the current state of the art. All seven models were released in 2025 and support from 128k to 10 million input tokens. We access GPT-4.1 (OpenAI, 2025) and DeepSeek V3 (DeepSeek et al., 2025) via Microsoft Azure, Mistral Small 3.1 (Mistral, 2025) via the Mistral API, and Gemini 2.0 Flash (Team Gemini, 2025) & Gemma 3 27b (Team Gemma et al., 2025) via the Google AI Studio API. We then host Qwen 3 32b (Yang et al., 2025)[10] and Llama 4 Scout (Meta, 2025) on two Nvidia H200 on AWS.[11] Values for each length bin are averaged over 10 novels.

Dev.	Model	Context	Release	OW
Meta	Llama 4 Scout	10,000,000	4/2025	✓
OpenAI	GPT-4.1	1,000,000	4/2025	×
DeepSeek	DeepSeek V3	1,000,000	2/2025	✓
Google	Gemini 2.0 Flash	1,000,000	2/2025	×
Google	Gemma 3 27b	128,000	4/2025	✓
Alibaba	Qwen 3 32b	128,000	4/2025	✓
Mistral	Mistral Small 3.1	128,000	3/2025	✓

Table 1: Comparison of recent large language models sorted by context window size and release date. OW indicates open weights.

Full-novel performance. We first report results where the model is asked to analyze the entire input text (whole unaltered novels, treatment 2 in Table 2). We find that all models exhibit comparable performance when processing volumes with <64k tokens but that performance begins to degrade as book lengths exceed 64k tokens. Performance degrades at different rates, with summary and storyworld scores dropping faster than time estimate scores. Open-weight models (particularly Gemma 3 27b and Qwen 3 32b) exhibit the steepest decline in performance. Of the models equipped to process over 128k tokens, we find GPT-4.1 is most consistent across all context lengths. Llama 4 Scout and Gemini 2.0 Flash are the next most resilient, achieving reliable performance in summary and time estimation over all lengths. However, no model performed well in estimating storyworlds, suggesting models grow increasingly inconsistent in their descriptions when processing individual chapters versus whole chapters. Finally, we find that performance scales with parameter count.

Treatment impact. Examining the effect of each treatment on average model performance reveals several trends. First, increasing novel length decreases model performance across all treatments (excluding title/author only, T1). Even when we only ask models to analyze a subset of the provided text (the "window of interest"), the same pattern

[5] We provide example summary responses in Appendix C.

[6] Note we pass in text author and title for all inputs independent of T1. This condition tests for text memorization.

[7] The mean chapter length in our corpus is 2,845 words or 3,696 Gemma 2 tokens.

[8] We concatenate chapter-level summaries; take the last recorded location of a character across all chapters (recurrently passing in characters from previous chapters to stabilize predictions); and sum per-chapter predictions in seconds.

[9] Semantic similarity for summaries, Jaccard similarity plus semantic similarity for storyworld descriptions, and absolute relative error for time.

[10] We disable chain-of-thought token sampling for Qwen 3 32b to maintain even footing with the other models.

[11] We consume a total of $600 in compute credits across all services.